

Chapter 3 exercises — Generating Functions

78 exercises, extracted from the textbook PDF.

Caveats. (1) Statements are auto-extracted; `fi` / `fl` ligatures dropped by the PDF text layer were repaired by a cleanup pass (a rare odd gap may remain). (2) A $\triangle$ marks exercises whose statement includes a **displayed formula** that did not survive extraction — read those in the book at the cited page. The book PDF is always authoritative.

#	p.	Exercise
3.1	96	Find the OGFs for each of the following sequences: $\{2k+1\}_{k \geq 0}$,
3.2	96	Find $[z^N]$ for each of the following OGFs: $1, (1-3z)^4$
3.3	96	Differentiate the OGF for harmonic numbers to verify the last line of Table 3.1.
3.4	96	$\triangle$ Prove that $\sum$
3.5	96	By factoring $1 - z^M \ln(1-z)^{M+1} / (1-z)$ in two different ways (and performing the associated convolutions), prove a general identity satisfied by the harmonic numbers and binomial coefficients.
3.6	96	Find the OGF for $\{\sum_{0 < k < n}$
3.7	96	Find the OGF for $\{H_k/k\}_{k \geq 1}$.
3.8	96	Find $[z^N]$ for each of the following OGFs: $1/(2(1-\ln(1-z)))$. Use the notation (2)
3.9	99	Find the EGFs for each of the following sequences: $\{2k+1\}_{k \geq 0}$,
3.10	99	Find the EGFs for $1, 3, 5, 7, \dots$ and $0, 2, 4, 6, \dots$
3.11	99	Find $N![z^N]A(z)$ for each of the following EGFs: $(2+1)z \ln z$, $A(z) = \ln z$, $A(z) = e^z + z$, $A(z) = 1/(1-z)$
3.12	99	Show that $\int_0^{\infty} z^{t-1} e^{-tz} N![z^N] e^z dt = H_N$. (Hint: Form a differential equation for the EGF $H(z) = \sum_{N \geq 0} H_N z^N/N!$.)
3.13	101	Given the EGF $A(z)$ for a sequence $\{a_k\}$, find the EGF for the sequence $\{\sum_{k=0}^N a_k\}$.
3.14	101	Given the EGF $A(z)$ for a sequence $\{a_k\}$, show that the OGF for the sequence is given by $\int_0^{\infty}$
3.15	104	Find the EGF for the Fibonacci numbers.
3.16	104	Use generating functions to solve the following recurrences: $a_n = -a_{n-1} + 6a_{n-2}$, $a_n = 11a_{n-2} - 6a_{n-3}$, $a_n = 3a_{n-1} - 4a_{n-2}$, $a_n = a_{n-1} - a_{n-2}$
3.17	108	Solve the recurrence $a_n = 5a_{n-1} - 8a_{n-2} + 4a_{n-3}$
3.18	108	Solve the recurrence $a_n = 2a_{n-2} - a_{n-4}$
3.19	108	Solve the recurrence $a_n = 6a_{n-1} - 12a_{n-2} + 18a_{n-3} - 27a_{n-4}$
3.20	108	Solve the recurrence $a_n = 3a_{n-1} - 3a_{n-2} + a_{n-3}$
3.21	108	Solve the recurrence $\sum_{k=1}^t a_n = -(-1)^k a_{n-k}$

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3.22	110	Use generating functions to solve the recurrence $na_n = (n - 2)a_{n-1} + 2$
3.23	110	[Greene and Knuth [6]] Solve the recurrence $na_n = (n + t - 1)a_{n-1}$
3.24	110	Solve the recurrence $t \sum a_n = n + 1 + a_{k-1} n$
3.25	112	Use Taylor's theorem to find the expansions of the following functions: $\sin(z)$,
3.26	112	Use Taylor's theorem to verify that the coefficients of the series expansion of $(1 - az - bz^2)^{-1}$ satisfy a second-order linear recurrence with constant coefficients.
3.27	112	Use Taylor's theorem to verify directly that $H(z) =$
3.28	112	Find an expression for $\frac{1}{[z^n]} \sqrt{\ln \frac{1-z}{1-z^2}}$ (Hint: Expand $(1 - z)^{-\alpha}$ and differentiate with respect to α .)
3.29	113	Find an expression for $[z^n]$
3.30	115	Show that $\sum (2k)(2N - 2k)k$
3.31	115	What recurrence on $\{C_N\}$ corresponds to multiplying both sides of the differential equation (3) for the quicksort generating function by $(1 - z)^2$?
3.32	116	Suppose that an OGF satisfies the differential equation $A'(z) = -A(z) +$
3.33	116	What identity on binomial coefficients is implied by the convolution $(1 + z)^r (1 - z)^s = (1 - z^2)^s (1 + z)^{r-s}$ where $r > s$?
3.34	116	Prove that $\sum (t - k)(k) 0 \leq k \leq t$
3.35	116	Use generating functions to evaluate
3.36	116	Use generating functions to find a sum expression for $[z^n]$
3.37	116	Use generating functions to find a sum expression for $[z^n] \cdot 2 - ez$
3.38	116	[Dobinski, cf. Comtet] Prove that $n![z^n] e^{-1} = e^{-1} z$
3.39	116	Prove the binomial transform identity using OGFs. Let $A(z)$ and $B(z)$ be related by $(z + 1)B(z) = A\left(\frac{z}{1-z}\right)$ and then use the change of variable $z = y/(y - 1)$.
3.40	116	Prove the binomial transform identity directly, without using generating functions. $\sum$

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3.41	116	[Faà di Bruno's formula, cf. Comtet] Let $f(z) = \sum_{n \geq 0} f_n z^n$ and $g(z) = \sum_{n \geq 0} g_n z^n$. Express $[z^k]f(g(z))$ using the multinomial theorem.
3.42	119	Show that the coefficients f_n in the expansion 2
3.43	119	Solve $f(z) = e^{-z} f$
3.44	119	Find an explicit formula for the OGF of the sequence satisfying the divide-and-conquer recurrence for $n > 1$ with $f_0 = 0$;
3.45	119	Iterate the following equation to obtain an explicit formula for $f(z)$: $f(z) = 1 + zf$
3.46	119	[Polya] Given $f(z)$ defined by the equation $f(z) =$
3.47	119	Prove that there is only one power series of the form $f(z) =$ that satisfies $f(z) = \sin(f(z))$.
3.48	119	Derive an underlying recurrence from the functional equation for 2-3 trees and use the recurrence to determine the number of 2-3 trees of 100 nodes.
3.49	122	Show that $(1 - z)^t C(t)(z) = \Psi(\Psi + 1) \dots (\Psi + t + 1)C(z)$.
3.50	122	Find the average number of exchanges used by median-of-three quicksort.
3.51	122	Find the number of comparisons and exchanges used, on the average, by quicksort when modified to use the median of v elements for partitioning.
3.52	123	[Euler] Discuss the solution of the differential equation $\sum_{0 \leq j \leq r}$
3.53	123	[van Emden, cf. Knuth] Show that, when the median of a sample of $2t+1$ elements is used for partitioning, the number of comparisons used by quicksort is $1/N \ln N + O(1/N)$. $H_{2t+2} - H_{t+1}$
3.54	126	Modify the above derivation to derive directly the generating function for the number of binary trees with N external nodes.
3.55	127	Discuss the form of an expression for $[z^N]D(z)$.
3.56	127	Write an efficient computer program that can compute $[z^N]D(z)$, given N .
3.57	127	Show that the generating function for the number of ways to express N as a linear combination (with integer coefficients) of powers of 2 is $\prod$
3.58	127	[Euler] Show that $1 = (1 + z)(1 + z^2)(1 + z^4)(1 + z^8) \dots 1 - z$ Give a closed form for the product of the first t factors. called the "computer

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		scientist's identity." Why?
3.59	127	Generalize the previous exercise to base 3.
3.60	127	Express $[z^N](1 - z)(1 - z^2)(1 - z^4)(1 - z^8) \cdots$ in terms of the binary representation of N .
3.61	130	Give a simple expression for $\text{var}(X)$ in terms of $r_k = \Pr\{X \leq k\}$.
3.62	130	Define $\text{mean}(P) \equiv P'(1)$ and $\text{var}(P) \equiv P''(1) + P'(1) - P'(1)^2$. Prove that $\text{mean}(PQ) = \text{mean}(P) + \text{mean}(Q)$ and $\text{var}(PQ) = \text{var}(P) + \text{var}(Q)$ for any differentiable functions P and Q with $P(1) = Q(1) = 1$, not just PGFs.
3.63	130	Verify the above results from the closed form $P_n(u) =$
3.64	131	Find the PGF for the random variable that counts the number of leading 0s in a random binary string, and use the PGF to find the mean and standard deviation.
3.65	136	Calculate the variance for the number of 1 bits in a random binary string of length n , using Table 3.5 and $p_n(u) = (1 + u)^n$, as shown earlier.
3.66	138	Verify from the vertical expansion that the mean for the binomial distribution is $n/2$ by first calculating $r_k(z)$ as described earlier.
3.67	139	Confirm that $q_{[1]}(z) \equiv$
3.68	139	Extract the coefficient of z^N in $q_{[2]}(z) + q_{[1]}(z)$ and verify the exact expression for the variance given in eorem 3.9. (See Exercise 3.8.)
3.69	144	Give closed-form expressions for the following: $\sum_{k=0}^n \binom{n}{k} z^k k!$
3.70	144	Prove from the generating function that $\sum_{k=0}^n \binom{n}{k} k^{n-1} = n \cdot 2^{n-1}$.
3.71	144	Prove from the generating function that $\sum_{k=0}^n \binom{n}{k} k^{n-1} = (n-1) \cdot 2^{n-1} + n$.
3.72	144	Prove from the generating function that $\sum_{k=0}^n \binom{n}{k} k^{n-1} = k \cdot 2^{n-1} + n$.
3.73	144	Prove that $B_m(0) = B_m(1) = B_m$ for all $m > 1$.
3.74	144	Prove from the generating function that B_k is zero for k odd, $k \geq 3$.
3.75	146	Show that, for any $k \geq 0$, the DGF for the characteristic sequence of numbers whose binary representation ends in k 0s is $\zeta(z)/2^{k+1}$.
3.76	146	Find the DGF for the function ψ_N , the number of trailing 0s in the binary representation of N .

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3.77	146	Find the DGF for the characteristic function of $\{N^2\}$.
3.78	146	$\triangle$ Prove that $\sum k$